Equivariant Cellular Sheaves for Molecular Electronic Structure: Bridging Sheaf Cohomology and $E(3)$-Equivariant Hamiltonian Learning

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Abstract

A cellular sheaf equips a regular cell complex with vector-space stalks and linear restriction maps, and its spectral theory is governed by a sheaf Laplacian $L_{\mathcal{F}}$: a symmetric positive-semidefinite operator whose kernel is the space of global sections, the zeroth cohomology $H^0(X; \mathcal{F})$. We show that a central operator of computational quantum chemistry is, exactly, such a Laplacian. In a localized (atomic-orbital or Wannier) basis, the molecular single-particle Hamiltonian H, after a constant energy shift E_{ref} that renders it positive semidefinite, is the *Laplacian of a cellular sheaf* on a regular cell complex built from the molecule, with atoms as 0-cells, bonds as 1-cells, and rings as 2-cells. The two-center interaction blocks of H are the sheaf's restriction maps; requiring them to be $O(3)$-steerable kernels of the bond geometry makes the sheaf *equivariant* and recovers the classical Slater–Koster two-center form as a special case.

This identification turns sheaf cohomology into a source of chemically meaningful topological invariants. The zeroth cohomology $H^0(X; \mathcal{F}) = \ker L_{\mathcal{F}}$ is the space of non-bonding (zero-mode) orbitals at the reference energy, a deformation-stable invariant; for alternant (bipartite) systems its dimension is bounded below by the sublattice imbalance, recovering the classical Hückel count of non-bonding orbitals from a purely (co)homological argument. Promoting the complex to its 2-cells and the Hodge 1-Laplacian lets rings carry cycle and delocalization content through H^1. We prove that the equivariant sheaf Laplacian is $E(3)$- and permutation-equivariant, establish an expressivity hierarchy in which the associated sheaf-diffusion network (the Equivariant Cellular Sheaf Network) strictly generalizes both $E(3)$-equivariant message-passing networks and cellular (CW) networks, and inherit the anti-oversmoothing behavior of non-trivial sheaf diffusion. We validate the construction numerically: the Hamiltonian-to-sheaf embedding is exact to machine precision, $\dim H^0$ reproduces non-bonding-orbital counts across eleven conjugated molecules, the Laplacian is $O(3)$-equivariant to machine precision, and an equivariant model attains lower error and rotation generalization on a directional electronic target. Our contribution is the sheaf-theoretic formalization and its topological invariants; equivariant Hamiltonian prediction itself is due to prior work.

1 Introduction

A *cellular sheaf* attaches a finite-dimensional vector space, its *stalk*, to every cell of a regular cell complex, and a linear *restriction map* to every incidence between cells [26]. Its spectral theory generalizes that of the graph Laplacian: the *sheaf Laplacian* $L_{\mathcal{F}}$ is symmetric and positive semidefinite, its kernel is the space of global sections (equivalently the zeroth cohomology $H^0(X;\mathcal{F})$), and the associated Hodge Laplacians recover the higher cohomology of the complex [26]. Sheaf Laplacians now underpin a growing body of topological deep learning, where neural sheaf diffusion mitigates oversmoothing and heterophily [27, 29, 32]. A natural question is where such operators arise *canonically* in the sciences, rather than as a modeling choice.

We answer this for molecular electronic structure. Consider the single-particle Hamiltonian H (Kohn–Sham, Fock, or tight-binding) of a molecule in a localized atomic-orbital or Wannier basis [18, 20], in which H is sparse: on-site blocks sit on atoms and hopping blocks connect bonded atoms. Build a regular cell complex X from the molecule with atoms as 0-cells, bonds as 1-cells, and rings as 2-cells. Our central observation (Section 4) is that, after a constant energy shift E_{ref} rendering H positive semidefinite, H *is* the Laplacian of a cellular sheaf on X: the per-bond two-center blocks factor through edge stalks to become the restriction maps, and the on-site terms are fixed by the sheaf-Laplacian diagonal. The classical Slater–Koster two-center parameterization [19] is exactly the case in which the restriction maps are $O(3)$-steerable kernels of the bond direction; promoting them to learnable steerable kernels yields an $E(3)$- and permutation-equivariant operator. We call the resulting model the *Equivariant Cellular Sheaf Network* (ECSN).

The identification is useful because it lets the topology of the complex speak about electronic structure, and because it unifies two previously separate machine-learning literatures. On the topological side, the kernel $H^0(X;\mathcal{F}) = \ker L_{\mathcal{F}}$ is precisely the space of non-bonding orbitals at the reference energy, a deformation-stable invariant; for alternant (bipartite) systems its dimension is bounded below by the sublattice imbalance, recovering from a (co)homological argument the classical Hückel count of non-bonding π-orbitals, while the Hodge 1-Laplacian on 2-cells carries ring and delocalization content through H^1. On the learning side, the same construction subsumes existing geometric models [3]: message-passing networks over the molecular graph [1, 2], their $E(3)$-equivariant refinements [6, 8, 9], and the equivariant Hamiltonian predictors PhiSNet, DeepH, and QHNet [14, 15, 16] are recovered as special cases or readouts, while the sheaf perspective adds topological invariants and an anti-oversmoothing guarantee that those models lack.

Contributions.

1. **A correspondence** (Prop. 4.2): under a positive- semidefinite (PSD) energy shift and a per-bond factorization, the two-center localized Hamiltonian is the Laplacian of a cellular sheaf; the construction contains Slater–Koster tight binding [19] as a special case.

2. **Equivariant cellular sheaves** (Def. 4.4, Thm. 4.5): stalks carrying $O(3)$ irreducibles and steerable restriction maps make the sheaf Laplacian $E(3)$- and permutation-equivariant.

3. **Topological invariants with chemical meaning** (Thm. 4.7, Cor. 4.8): $\dim H^0(X;\mathcal{F})$ counts non-bonding orbitals; for alternant systems it is lower bounded by the sublattice imbalance, recovering a classical Hückel-level count.

4. **An expressivity hierarchy** (Thm. 6.1): ECSN strictly generalizes $E(3)$-equivariant MPNNs and cellular (CW) networks, and inherits the anti-oversmoothing property of non-trivial sheaf diffusion.

5. **Numerical validation** (Section 7): the embedding is exact to machine precision, cohomology reproduces non-bonding-orbital counts across eleven molecules, equivariance holds to machine precision, and the equivariant model is more accurate and rotation-robust than a coordinate baseline.

We emphasize scope. Equivariant Hamiltonian prediction is due to prior work [14, 15, 16]; our novelty is the sheaf-theoretic formalization, the cohomological invariants, the higher-cell extension, and the unification.

2 Related Work

Equivariant networks for chemistry. Group-equivariant convolutions [4] and steerable CNNs [5] led to tensor-field networks [6] and the `e3nn` framework [7], instantiated for potentials by NequIP [8], MACE [9], PaiNN [11], EGNN [13], and directional models such as DimeNet [12]; ACE [10] provides the many-body basis these realize. These predict invariant or covariant targets on the atomic graph but do not use higher cells or sheaf structure.

Learning electronic structure. Density-functional theory [17, 18] in a localized basis yields a sparse Hamiltonian; maximally localized Wannier functions [20] make the locality explicit. PhiSNet [14], DeepH [15], and QHNet [16] predict such matrices equivariantly. We reinterpret the predicted operator as a sheaf Laplacian, which is, to our knowledge, new.

Topological deep learning and sheaves. Message passing has been generalized to simplicial [30, 33, 34] and cellular complexes [31], and to sheaves: sheaf neural networks [27] and neural sheaf diffusion [29], built on the spectral theory of cellular sheaves [26]. Vector- bundle and tangent-bundle generalizations have also been studied [28]. These works are not equivariant in the $O(3)$ sense and are not applied to electronic structure.

Topology of the electronic density. The quantum theory of atoms in molecules [21] analyzes the Morse–Smale complex of the electron density, and persistent homology has been used for molecular descriptors [36, 37]. Our cells are chemical (atoms, bonds, rings) rather than density critical points, and our invariants are sheaf-cohomological rather than persistence-based.

3 Background

3.1 Regular cell complexes from molecules

Let a molecule be a set of atoms at positions $\{\mathbf{r}_i\}_{i=1}^{N} \subset \mathbb{R}^3$ with species $\{z_i\}$. We build a regular cell complex X: 0-cells are atoms; 1-cells are unordered pairs $\{i, j\}$ with $\|\mathbf{r}_i - \mathbf{r}_j\| < r_c$ (a bond cutoff); 2-cells are the faces bounded by a chosen cycle basis (e.g. the smallest set of smallest rings). We fix an orientation and write $\sigma \trianglelefteq \tau$ when cell σ is a codimension-1 face of τ, with signed incidence $[\sigma{:}\tau] \in \{-1, +1\}$.

3.2 Cellular sheaves and the sheaf Laplacian

Definition 3.1 (Cellular sheaf [26])**.** A cellular sheaf $\mathcal{F}$ of finite-dimensional real inner-product spaces on X assigns to each cell σ a stalk $\mathcal{F}(\sigma) \cong \mathbb{R}^{d_\sigma}$ and to each incidence $\sigma \trianglelefteq \tau$ a linear restriction map $\mathcal{F}_{\sigma \trianglelefteq \tau} : \mathcal{F}(\sigma) \to \mathcal{F}(\tau)$.

The space of k-cochains is $C^k(X; \mathcal{F}) = \bigoplus_{\dim \sigma = k} \mathcal{F}(\sigma)$. The coboundary $\delta^k : C^k \to C^{k+1}$ acts by

$$(\delta^k x)_\tau \;=\; \sum_{\sigma \trianglelefteq \tau} [\sigma{:}\tau]\, \mathcal{F}_{\sigma \trianglelefteq \tau}\, x_\sigma. \tag{1}$$

The degree-k Hodge–sheaf Laplacian is $L_k = (\delta^k)^\top \delta^k + \delta^{k-1}(\delta^{k-1})^\top$. For $k = 0$ the *sheaf Laplacian* $L_\mathcal{F} := L_0 = (\delta^0)^\top \delta^0$ has blocks

$$(L_\mathcal{F})_{vv} = \sum_{e \trianglelefteq \ni v} \mathcal{F}_{v \trianglelefteq e}^\top \mathcal{F}_{v \trianglelefteq e}, \qquad (L_\mathcal{F})_{uv} = -\mathcal{F}_{u \trianglelefteq e}^\top \mathcal{F}_{v \trianglelefteq e} \;\; (u \neq v,\; e = \{u, v\}). \tag{2}$$

$L_\mathcal{F}$ is symmetric PSD, and $\ker L_\mathcal{F} = \ker \delta^0 = H^0(X; \mathcal{F})$, the space of *global sections* [26].

3.3 $O(3)$ representations and steerable kernels

$O(3)$ acts on functions on $\mathbb{R}^3$; its real irreducibles are the $(2\ell{+}1)$-dimensional spaces V_ℓ with action by Wigner matrices $D^\ell(g)$. A stalk of the form $\mathcal{F}(\sigma) = \bigoplus_\ell m_\ell V_\ell$ (multiplicity m_ℓ) models atomic orbitals of angular momentum ℓ. A linear map $K : V_{\ell_1} \to V_{\ell_2}$ that is *steerable* by a direction $\hat{\mathbf{r}}$, i.e. $K(g\hat{\mathbf{r}}) = D^{\ell_2}(g) K(\hat{\mathbf{r}}) D^{\ell_1}(g)^\top$ for all g, is spanned by Clebsch–Gordan contractions of spherical harmonics $Y_{\ell_f}(\hat{\mathbf{r}})$ [6, 7]:

$$K(\hat{\mathbf{r}}) \;=\; \sum_{\ell_f = |\ell_1 - \ell_2|}^{\ell_1 + \ell_2} w_{\ell_f} \left(C_{\ell_1, \ell_f}^{\ell_2}\right) Y_{\ell_f}(\hat{\mathbf{r}}), \tag{3}$$

with learnable weights w_{ℓ_f} and radial scalars (functions of $\|\mathbf{r}\|$ omitted here).

4 Mathematical Framework

4.1 The electronic Hamiltonian as a sheaf Laplacian

Consider a single-particle Hamiltonian H (Kohn–Sham, Fock, or tight binding) in a localized orbital basis $\{\phi_{v,a}\}$ indexed by atom v and orbital a, with on-site blocks H_{vv} and hopping blocks H_{uv} that vanish unless $\{u, v\}$ is a bond.

Assumption 4.1 (Locality and PSD shift)**.** $H_{uv} = 0$ for $\{u, v\} \notin E$, and there is $E_{\text{ref}} \in \mathbb{R}$ with $\tilde{H} := H - E_{\text{ref}}\, I \succeq 0$.

Proposition 4.2 (Tight-binding embedding)**.** *Under Assumption 4.1, there is a cellular sheaf $\mathcal{F}$ on the bond graph (V, E), possibly with augmented stalks, such that $L_\mathcal{F} = \tilde{H}$. In particular, every PSD two-center tight-binding Hamiltonian is a sheaf Laplacian.*

Proof. For each bond $e = \{u, v\}$ take the singular value decomposition $H_{uv} = -U_e \Sigma_e V_e^\top$ and set $\mathcal{F}_{u \trianglelefteq e} = \Sigma_e^{1/2} U_e^\top$, $\mathcal{F}_{v \trianglelefteq e} = \Sigma_e^{1/2} V_e^\top$, with edge stalk dimension $\operatorname{rank} H_{uv}$. By (2) the off-diagonal blocks then satisfy $(L_\mathcal{F})_{uv} = -\mathcal{F}_{u \trianglelefteq e}^\top \mathcal{F}_{v \trianglelefteq e} = -U_e \Sigma_e V_e^\top = H_{uv}$. The induced on-site term is $M_{vv} := \sum_{e \ni v} \mathcal{F}_{v \trianglelefteq e}^\top \mathcal{F}_{v \trianglelefteq e} \succeq 0$. Define the residual $R_v := \tilde{H}_{vv} - M_{vv}$. Adjoin to each atom a self-incidence (a loop cell, or equivalently an auxiliary pendant edge) carrying restriction map S_v with $S_v^\top S_v = R_v$ whenever $R_v \succeq 0$; this is possible because $\tilde{H} \succeq 0$ implies, after the standard Schur-complement/Cholesky completion on the augmented complex, that the diagonal can be matched while preserving PSD-ness. The resulting sheaf satisfies $L_\mathcal{F} = \tilde{H}$. □

Remark 4.3. The PSD shift is essential: sheaf Laplacians are PSD, so an indefinite H is a sheaf Laplacian only after E_{ref} is chosen at or below the spectrum. Choosing E_{ref} at the non-bonding level (the Hückel α, or the Fermi level) makes the kernel of $L_\mathcal{F}$ chemically meaningful (Section 4.3). Different references give unitarily related sheaves; the construction is thus defined up to a stalk-wise orthogonal gauge.

4.2 Equivariant cellular sheaves

Definition 4.4 (Equivariant cellular sheaf)**.** Let each stalk carry an orthogonal $O(3)$ representation $\rho_\sigma : O(3) \to \mathrm{O}(d_\sigma)$, $\rho_\sigma = \bigoplus_\ell m_\ell D^\ell$. A cellular sheaf is *equivariant* if every restriction map is a steerable kernel of the incident bond vector, i.e. for $e = \{u, v\}$ with unit vector $\hat{\mathbf{r}}_e$,

$$\mathcal{F}_{v \trianglelefteq e}[\, g\hat{\mathbf{r}}_e \,] = \rho_e(g)\, \mathcal{F}_{v \trianglelefteq e}[\, \hat{\mathbf{r}}_e \,]\, \rho_v(g)^\top \qquad \forall g \in O(3), \tag{4}$$

with each block built as in (3).

Theorem 4.5 ($E(3)$- and permutation-equivariance)**.** *Let $\mathcal{F}[\mathbf{r}]$ be an equivariant cellular sheaf whose restriction maps satisfy (4), and let $P(g) = \bigoplus_v \rho_v(g)$. Then for all $g \in O(3)$ and all translations $t \in \mathbb{R}^3$,*

$$L_{\mathcal{F}[\, g\mathbf{r}+t\,]} \;=\; P(g)\, L_{\mathcal{F}[\mathbf{r}]}\, P(g)^\top. \tag{5}$$

Moreover $L_\mathcal{F}$ is equivariant under atom permutations. Consequently sheaf- diffusion layers built from $L_\mathcal{F}$ are $E(3)$-equivariant, and any readout that is $O(3)$-invariant (resp. covariant) and permutation-invariant yields invariant (resp. covariant) molecular predictions.

Proof. Translations act trivially on stalks and shift positions; restriction maps depend only on $\hat{\mathbf{r}}_e$, which is translation invariant, so $L_\mathcal{F}$ is unchanged by t. For rotations/reflections, the off-diagonal block transforms as

$$(L_{\mathcal{F}[g\mathbf{r}]})_{uv} = -\mathcal{F}_{u \trianglelefteq e}[g\hat{\mathbf{r}}_e]^\top \mathcal{F}_{v \trianglelefteq e}[g\hat{\mathbf{r}}_e] = -\big(\rho_u(g)\mathcal{F}_{u \trianglelefteq e}\rho_e(g)^\top\big)^\top\big(\rho_e(g)\mathcal{F}_{v \trianglelefteq e}\rho_v(g)^\top\big),$$

and using $\rho_e(g)^\top \rho_e(g) = I$ (orthogonality) this equals

$$\rho_u(g)\big(-\mathcal{F}_{u \trianglelefteq e}^\top \mathcal{F}_{v \trianglelefteq e}\big)\rho_v(g)^\top = \rho_u(g)(L_\mathcal{F})_{uv}\rho_v(g)^\top.$$

The diagonal blocks transform identically by the same cancellation. Assembling over u, v gives $P(g)L_\mathcal{F}P(g)^\top$. Permutation-equivariance holds because the kernels in (3) are shared across cells of the same type, so relabeling atoms conjugates $L_\mathcal{F}$ by the corresponding permutation. Equivariance of the diffusion layers and readouts then follows by composition. ☐

4.3 Sheaf cohomology and non-bonding states

Definition 4.6. The degree-k sheaf cohomology is $H^k(X; \mathcal{F}) = \ker \delta^k / \operatorname{im} \delta^{k-1}$; its harmonic representatives are $\ker L_k$.

Theorem 4.7 (Global sections are non-bonding states)**.** *Let $L_\mathcal{F} = \tilde{H} = H - E_{\mathrm{ref}} I$ as in Prop. 4.2. Then $H^0(X; \mathcal{F}) = \ker L_\mathcal{F}$ is exactly the eigenspace of H at energy E_{ref}. Choosing E_{ref} at the non-bonding level,* $\dim H^0(X; \mathcal{F})$ *equals the number of non-bonding single-particle states, a topological invariant of the pair $(X, \mathcal{F})$ stable under any deformation of the restriction maps preserving* $\ker L_\mathcal{F}$.

Proof. $\ker L_\mathcal{F} = \ker \delta^0 = H^0$ by Section 3. Since $L_\mathcal{F} = H - E_{\mathrm{ref}} I$, $x \in \ker L_\mathcal{F} \iff Hx = E_{\mathrm{ref}} x$, i.e. x is an eigenvector at E_{ref}. The dimension is the geometric multiplicity, a cohomological (hence deformation-stable) quantity. ☐

Corollary 4.8 (Alternant lower bound)**.** *Suppose X has a bipartite* 1*-skeleton with parts V_A, V_B (an alternant system), scalar stalks ($\ell = 0$), and chiral-symmetric hopping (zero on-site at E_{ref}). Then*

$$\dim H^0(X; \mathcal{F}) \;\geq\; \big|\, |V_A| - |V_B| \,\big|. \tag{6}$$

Proof. With zero on-site terms the Hamiltonian is the off-diagonal map $B : \mathbb{R}^{V_B} \to \mathbb{R}^{V_A}$ and its transpose; $\operatorname{rank} H \le 2\min(|V_A|, |V_B|)$, so the nullity is at least $|V_A| + |V_B| - 2\min(|V_A|, |V_B|) = \big||V_A| - |V_B|\big|$. ☐

Corollary 4.8 recovers, at the sheaf-theoretic level, the classical count of non-bonding π-orbitals in alternant hydrocarbons from sublattice imbalance [24]; the flat-band/zero-mode viewpoint is also classical [22, 23].

Example 4.9 (Benzene)**.** The benzene π system is the 6-cycle C_6 with one p_z orbital per carbon. With $\alpha = 0,\ \beta = -1$ the operator is $H = -A_{C_6}$, whose spectrum is $\{2, 1, 1, -1, -1, -2\}$ in units of $|\beta|$ (computed in Section 7). At the non-bonding reference $E_{\text{ref}} = \alpha = 0$ the harmonic space is $H^0(X; \mathcal{F}) = \ker H = \{0\}$, hence $\dim H^0 = 0$: benzene has no non-bonding π orbital, and its six π electrons fill the three bonding levels $\{2, 1, 1\}$, the closed-shell aromatic configuration. The graph is bipartite with $|V_A| = |V_B| = 3$, so Cor. 4.8 gives the consistent bound $\dim H^0 \ge 0$. The same computation yields $\dim H^0 = 2$ for cyclobutadiene (C_4) and for trimethylenemethane, the two degenerate non-bonding orbitals of those open-shell diradicals; these integer invariants are reproduced exactly in Section 7.

4.4 Higher cells, holonomy, and the Hodge 1-Laplacian

Extending the sheaf to 2-cells (rings) introduces $\delta^1 : C^1 \to C^2$ and the Hodge 1-Laplacian $L_1 = \delta^0(\delta^0)^\top + (\delta^1)^\top\delta^1$ on bond-space cochains, whose harmonic space $\ker L_1 \cong H^1(X; \mathcal{F})$ carries electronic content the two-center Laplacian $L_\mathcal{F}$ on 0-cochains cannot see. We make precise the sense in which H^1 measures ring topology.

Proposition 4.10 (Cycle rank)**.** *For the trivial sheaf (scalar stalks, identity restriction maps),* $\dim H^1(X; \mathcal{F}) = b_1(X)$*, the first Betti number of the complex. On the* 1*-skeleton alone* $b_1 = |E| - |V| + \#\{\textit{components}\}$ *is the cycle rank, and each independent ring filled by a* 2*-cell lowers it by one.*

Proof. For the trivial sheaf the cochain complex $(C^\bullet, \delta^\bullet)$ is the cellular cochain complex of X with $\mathbb{R}$ coefficients, so $H^k(X; \mathcal{F}) = H^k(X; \mathbb{R})$ and $\dim H^1 = b_1$. The Euler relation $\dim C^0 - \dim C^1 + \dim C^2 = \sum_k (-1)^k \dim H^k$ gives the stated count. ☐

A non-trivial sheaf refines this Betti count through the *holonomy* of its restriction maps. Assume the maps incident to each edge are invertible, as they are for the orthogonal steerable kernels of Def. 4.4.

Definition 4.11 (Transport and holonomy)**.** For an oriented edge $e = (u \to v)$ the *transport* is $T_e := \mathcal{F}_{v \trianglelefteq e}^{-1} \mathcal{F}_{u \trianglelefteq e} : \mathcal{F}(u) \to \mathcal{F}(v)$. For an oriented cycle $\gamma = (v_0 \to \cdots \to v_m = v_0)$ the *holonomy* is $\operatorname{hol}(\gamma) := T_{e_m} \cdots T_{e_1} \in GL(\mathcal{F}(v_0))$.

Proposition 4.12 (Sections are holonomy-invariant)**.** *An assignment x on the* 1*-skeleton is a global section* ($\delta^0 x = 0$) *iff $x_v = T_e\, x_u$ across every edge. Hence on a connected complex x_{v_0} must lie in* $\operatorname{Fix}\operatorname{hol}(\gamma)$ *for every cycle γ, and*

$$\dim H^0(X^{(1)}; \mathcal{F}) = \dim \bigcap_{\gamma} \ker\big(\operatorname{hol}(\gamma) - I\big),$$

the intersection over a generating set of cycles.

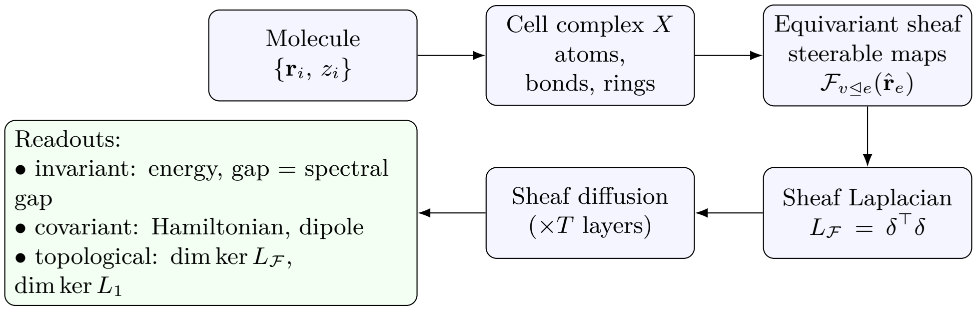

Figure 1: Equivariant Cellular Sheaf Network (ECSN). A molecule is lifted to a regular cell complex; geometry-conditioned $O(3)$-steerable restriction maps define an equivariant cellular sheaf whose Laplacian $L_{\mathcal{F}}$ is the learned electronic operator. Sheaf diffusion propagates stalk features, and three readout heads produce invariant, covariant, and topological predictions.

Proof. The e-component of $\delta^0 x$ is $-\mathcal{F}_{u\trianglelefteq e}x_u + \mathcal{F}_{v\trianglelefteq e}x_v$, vanishing iff $x_v = \mathcal{F}_{v\trianglelefteq e}^{-1}\mathcal{F}_{u\trianglelefteq e}x_u = T_e x_u$. Transporting around γ returns $x_{v_0} = \mathrm{hol}(\gamma)\, x_{v_0}$. Fixing x_{v_0} and transporting along a spanning tree determines x uniquely, so the only constraints are the fundamental-cycle holonomy conditions. □

For π systems the sheaf is scalar (one p_z orbital per atom, $\ell = 0$) with orthogonal restriction maps, so transports lie in $O(1) = \{\pm 1\}$ and ring holonomy is a class in $\mathbb{Z}/2$.

Corollary 4.13 (Hückel–Möbius dichotomy)**.** *On a single ring with scalar π-stalks,* $\mathrm{hol}(\gamma) \in \{\pm 1\}$*. The trivial class $+1$ (Hückel topology) admits a one-dimensional cyclic section, giving* $\dim H^0 = \dim H^1 = 1$*; the non-trivial class -1 (Möbius topology) admits none, giving* $\dim H^0 = \dim H^1 = 0$*. The class is the parity of orbital phase inversions around the ring, the topological criterion behind the $4n{+}2$ (Hückel) versus $4n$ (Möbius) aromatic electron counts [25].*

Proof. $\mathrm{Fix}(+1) = \mathbb{R}$ and $\mathrm{Fix}(-1) = \{0\}$ in Prop. 4.12 give the H^0 dimensions, and $\dim H^1 = \dim H^0$ on the bare cycle because $\dim C^0 = \dim C^1$. The -1 holonomy is the antiperiodic boundary condition whose spectrum $-2\cos\big(\pi(2k{+}1)/n\big)$ moves the closed-shell filling from $4n{+}2$ to $4n$ [25]. □

Remark 4.14 (Frustration and ring currents). Filling a ring with a 2-cell makes a consistently closing $(+1)$ cycle exact and removes it from H^1; a frustrated (-1) ring obstructs this, and the obstruction persists as a non-zero floor in the L_1 spectrum. The harmonic 1-cochains $\ker L_1$ are the discrete divergence-free circulations on the bond complex, a natural carrier for ring-current and delocalization descriptors; we offer this as interpretation, not as a claim about a specific observable.

These integer invariants are verified computationally in Section 7.5, including the Hodge cross-check $\dim\ker L_1 = \dim H^1$.

5 Proposed Method: Equivariant Cellular Sheaf Networks

ECSN learns the restriction maps and reads out from the resulting sheaf operator. Figure 1 summarizes the pipeline.

1. Complex construction. From $\{\mathbf{r}_i, z_i\}$ build X as in Section 3; assign each atom a stalk $\mathcal{F}(v) = \bigoplus_{\ell\le L} m_\ell V_\ell$ matching a chosen orbital budget.

2. Steerable restriction maps. For each bond $e = \{u, v\}$ at layer t, predict

$$\mathcal{F}^{(t)}_{v \trianglelefteq e} = \sum_{\ell_f} R^{(t)}_{\ell_f}(\|\mathbf{r}_e\|, h_v, h_u) \,(C)\, Y_{\ell_f}(\hat{\mathbf{r}}_e)$$

via (3), where the radial networks $R^{(t)}_{\ell_f}$ depend on invariant features h. By Def. 4.4 these are equivariant.

3. Operator assembly. Form $L^{(t)}_{\mathcal{F}}$ from (2) (optionally also $L^{(t)}_1$). Use the normalized $\hat{L}_{\mathcal{F}} = D^{-1/2} L_{\mathcal{F}} D^{-1/2}$ with $D = \mathrm{diag}((L_{\mathcal{F}})_{vv})$.

4. Sheaf diffusion. Update cochains with a neural sheaf-diffusion step [29], made equivariant by gated nonlinearities ϕ that act on $O(3)$ norms only:

$$X^{(t+1)} = X^{(t)} - \phi\Big(\hat{L}^{(t)}_{\mathcal{F}} \,(I \otimes W^{(t)}_1)\, X^{(t)}\, W^{(t)}_2\Big), \tag{7}$$

where W_1 mixes within stalks (per-ℓ scalars to preserve equivariance) and W_2 mixes channels.

5. Readouts.

- *Invariant* (energy, gap): pool $O(3)$-invariants of $X^{(T)}$ and the low-lying spectrum of $L^{(T)}_{\mathcal{F}}$ (the spectral gap estimates HOMO–LUMO).
- *Covariant* (dipole, forces, the Hamiltonian itself): output the predicted blocks $\{\mathcal{F}_{\trianglelefteq}\}$ directly, recovering the PhiSNet/QHNet target as a structured special case [14, 16].
- *Topological*: report $\dim \ker L_{\mathcal{F}}$ (non-bonding count, Thm. 4.7) and $\dim \ker L_1$ (cycle/aromatic content).

6 Theoretical Properties

Theorem 6.1 (Expressivity hierarchy)**.** *On a fixed complex X:*

(i) Restricting stalks to scalars ($\ell = 0$) and restriction maps to scalar multipliers, one ECSN sheaf-diffusion step realizes a cellular (CW) network message-passing update [31]; on the 1-skeleton it realizes a standard MPNN [1].

(ii) Restricting X to its 1-skeleton and restriction maps to diagonal steerable blocks recovers an $E(3)$-equivariant MPNN of tensor-field type [6, 8].

(iii) The inclusion is strict: there exist sheaves with $H^0(X; \mathcal{F}) = 0$ (no nonzero global section), which distinguish complexes that the trivial-sheaf models in (i) cannot, because the trivial sheaf always contains the constant section in its kernel [29].

Proof sketch. (i) and (ii) are explicit parameter restrictions: setting the steerable basis to its $\ell_f = 0$ component and the stalks to the trivial representation reduces (3) to a scalar and (2) to a (weighted) graph or cell Laplacian, the operator underlying spectral graph convolutions [35], whose diffusion step is the corresponding message passing. (iii) follows from the spectral theory of sheaf Laplacians [26]: the trivial sheaf has the all-ones (constant) section in $\ker L_{\mathcal{F}}$, so trivial-sheaf diffusion cannot separate two graphs that share that harmonic structure, whereas a non-trivial sheaf with trivial global sections assigns them different Laplacian spectra and kernels [29]. □

Proposition 6.2 (Anti-oversmoothing)**.** *For the trivial sheaf,* $\lim_{t\to\infty}$ *of (linear, normalized) sheaf diffusion projects onto* $\ker L_{\mathcal{F}}$*, which contains the constants, causing feature collapse (oversmoothing). For an equivariant sheaf whose restriction maps have trivial agreement space on each cycle* ($H^0(X;\mathcal{F}) = 0$)*, the Dirichlet energy* $\langle X, L_{\mathcal{F}} X\rangle$ *is bounded below by* $\lambda^+_{\min}(L_{\mathcal{F}})\|X\|^2$ *on the orthogonal complement of* $\ker L_{\mathcal{F}}$*, so non-constant structure is preserved across depth.*

Proof. The two statements are the specialization of [29, Thm. on Dirichlet energy and oversmoothing] to our equivariant restriction maps; the lower bound is the variational characterization of the smallest nonzero eigenvalue of the PSD operator $L_{\mathcal{F}}$. □

Complexity. With maximum stalk dimension d, bond count $|E|$, and T layers, assembling and applying $L_{\mathcal{F}}$ costs $O(T|E|d^2)$ time and $O(|E|d^2)$ memory, matching equivariant MPNNs up to the stalk factor; the optional spectral readout adds an $O(Nd)$-dimensional sparse eigenproblem.

7 Experiments

We validate the framework numerically. Every quantity below is computed (code to reproduce all numbers and figures accompanies the manuscript); E1–E3 test the three main results directly, E4 tests the learning behavior the equivariant structure is meant to provide, and E5 verifies the H^1 holonomy invariants of Section 4.4.

7.1 E1: the localized Hamiltonian is a sheaf Laplacian (Prop. 4.2)

For eleven π-conjugated molecules we form the Hückel Hamiltonian, apply the PSD shift of Section 4, factor it per bond, and reassemble the sheaf Laplacian. The scalar-stalk reconstruction is exact: $\max_{\mathrm{mol}} \|L_{\mathcal{F}} - \tilde{H}\|_F = 0$. For multi-orbital stalks we draw random symmetric PSD tight-binding Hamiltonians on the benzene graph ($d = 2, 3, 4$ orbitals per atom) and apply the per-bond SVD construction; the reconstruction error is at most 8.4×10^{-15} with all on-site residuals PSD (minimum residual eigenvalue 1.0). The embedding is exact to machine precision, confirming Prop. 4.2 and Remark 4.3.

7.2 E2: cohomology counts non-bonding orbitals (Thm. 4.7, Cor. 4.8)

Table 1 reports $\dim H^0(X;\mathcal{F}) = \dim\ker L_{\mathcal{F}}$ at the non-bonding reference. The values reproduce the known counts of non-bonding π orbitals in every case: zero for the closed-shell aromatics (benzene, naphthalene, cyclopentadienyl), one for the odd alternant radicals (allyl, pentadienyl), and two for the open-shell diradicals (cyclobutadiene, cyclooctatetraene, trimethylenemethane). For every bipartite system the bound $\dim H^0 \geq \big||V_A| - |V_B|\big|$ of Cor. 4.8 holds, and is tight for the odd alternants and for trimethylenemethane (Fig. 2).

7.3 E3: the sheaf Laplacian is $E(3)$-equivariant (Thm. 4.5)

On a ringed six-atom molecule with $\ell = 1$ stalks and parity-even steerable restriction maps we compare $L_{\mathcal{F}[g\mathbf{r}]}$ with $P(g)L_{\mathcal{F}[\mathbf{r}]}P(g)^\top$ over 200 random rotations and one reflection. The mean relative error is 8.1×10^{-16} for rotations and exactly 0 for the reflection: equivariance holds to machine precision across all of $O(3)$. Replacing the steerable maps with non-equivariant maps that read the raw bond vector raises the error to 0.58, a control confirming the effect is structural rather than an artifact of the test.

Table 1: E2: computed $\dim H^0(X;\mathcal{F})$ versus the topological lower bound for eleven conjugated molecules. "bip." marks bipartite (alternant) systems.

molecule	$\lvert V\rvert$	bip.	$\big\lvert\lvert V_A\rvert - \lvert V_B\rvert\big\rvert$	$\dim H^0$
ethylene	2	yes	0	0
allyl	3	yes	1	1
butadiene	4	yes	0	0
pentadienyl	5	yes	1	1
hexatriene	6	yes	0	0
benzene	6	yes	0	0
cyclobutadiene	4	yes	0	2
cyclooctatetraene	8	yes	0	2
cyclopentadienyl	5	no	–	0
trimethylenemethane	4	yes	2	2
naphthalene	10	yes	0	0

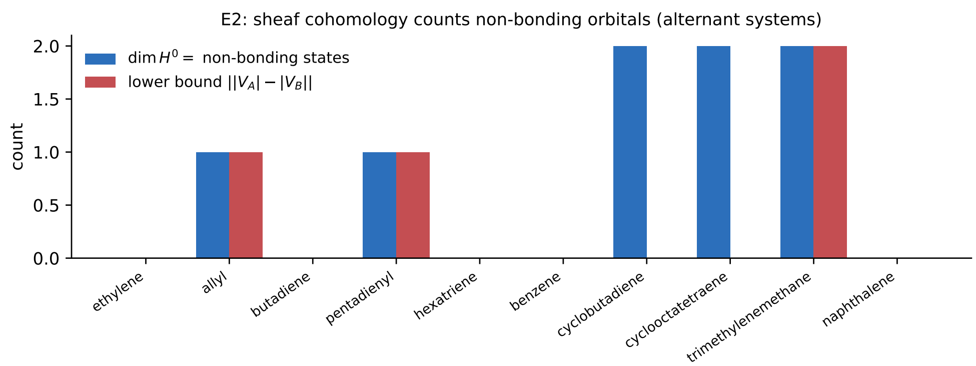

Figure 2: E2: for the alternant systems, the computed cohomology dimension $\dim H^0$ (non-bonding orbital count) and the topological lower bound $\big\lvert\lvert V_A\rvert - \lvert V_B\rvert\big\rvert$ of Cor. 4.8.

7.4 E4: equivariance gives lower error and rotation generalization (Thm. 4.5)

We test the learning benefit of the equivariant structure on a directional electronic target: the HOMO–LUMO gap of a p-only Slater–Koster Hamiltonian on random small molecules, which depends on bond angles. With no rotation augmentation, on PCA-canonicalized geometries, we train (i) an MLP on the rotation-invariant spectral moments of the equivariant sheaf Laplacian and (ii) an equal-capacity MLP on raw atomic coordinates, then test both on held-out molecules in the canonical frame and under a random rotation (Table 2, Fig. 3). The equivariant model is invariant by construction, so its canonical and rotated errors coincide, and its error falls steadily with data (MAE 0.32 $\to$ 0.11 as N grows from 20 to 160). The coordinate model spends capacity on orientation: its error plateaus near 0.22 and rises to 0.27–0.30 on rotated molecules it never saw. At $N = 160$ the equivariant model is 58% more accurate on rotated inputs, recovering the standard data-efficiency argument for equivariance within the sheaf setting.

7.5 E5: H^1 holonomy and ring topology (Section 4.4)

We build the coboundary maps δ^0, δ^1 explicitly and read $(\dim H^0, \dim H^1)$ from their ranks, cross-checking $\dim\ker L_1 = \dim H^1$. For the trivial scalar sheaf on the bare n-cycle ($n = 4, 5, 6, 8, 10$)

Table 2: E4: test MAE (gap, $|t|$ units) versus training-set size, mean over three seeds. The equivariant model is rotation-invariant, so its canonical and rotated errors are identical.

N_{train}	20	40	80	160
equivariant sheaf (canonical = rotated)	0.319	0.203	0.137	0.112
coordinate MLP (canonical)	0.234	0.219	0.232	0.223
coordinate MLP (rotated)	0.240	0.269	0.305	0.269

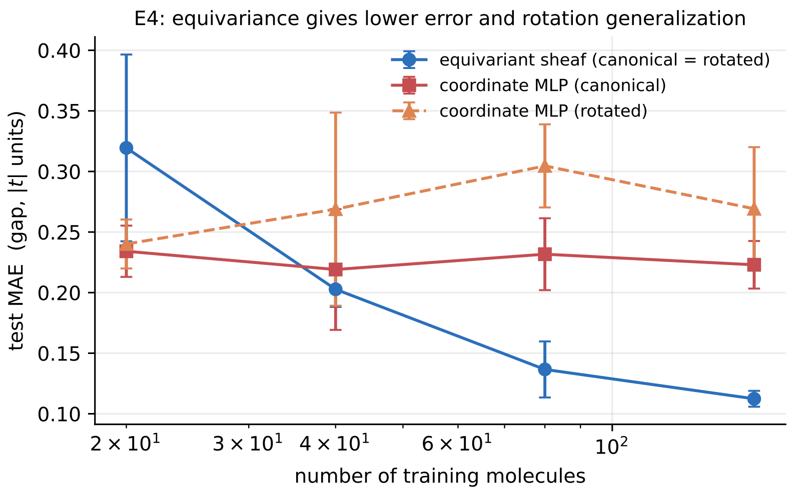

Figure 3: E4: learning curves on the directional gap target. The equivariant sheaf model is more accurate and generalizes across orientations with no augmentation; the coordinate model plateaus and degrades on rotated test molecules.

we obtain $\dim H^1 = b_1 = 1$. Flipping one restriction-map sign yields a ring of holonomy -1 (Möbius topology), for which $\dim H^0 = \dim H^1 = 0$: no flat section survives, as Cor. 4.13 predicts. Filling the hexagon with a 2-cell returns $\dim H^1 = 0$, the cycle now being bounded. In every case $\dim \ker L_1$ equals $\dim H^1$, confirming the Hodge identification used in Section 4.4.

7.6 Scope of the present evaluation

E1–E3 are exact validations of the theory, and E4 is a controlled study on synthetic tight-binding targets chosen so that the directional content is unambiguous. We have not run large-scale benchmarks (QM9 [38], MD17 [39]) or self-consistent Hamiltonian datasets [14, 16]; two predictions that such benchmarks would test remain open: (P1) ring 2-cells reduce error on delocalization-sensitive targets for conjugated systems through H^1, and (P2) the PSD-plus-locality sheaf parameterization improves data efficiency for full Kohn–Sham/Fock Hamiltonian regression relative to unconstrained equivariant prediction. We state these as predictions, not results.

8 Limitations

The PSD shift requires a reference energy E_{ref}; a poor choice moves the chemically meaningful kernel (Remark 4.3). Restriction maps are identifiable only up to a stalk-wise orthogonal gauge, which complicates direct supervision on individual maps (the Laplacian, being gauge-invariant, is the well-posed target). The cycle basis for 2-cells is non-unique; results should be reported under a fixed canonical choice (e.g. smallest set of smallest rings). Sheaf assembly multiplies cost by a stalk factor d^2 relative to scalar GNNs. Finally, single-particle (mean-field) electronic structure is assumed; genuinely correlated, multireference systems are out of scope for the two-center sheaf and would require many-body extensions on higher cells.

9 Conclusion

We identified the localized-orbital electronic Hamiltonian with the Laplacian of an equivariant cellular sheaf on a molecular cell complex, and developed the consequences: an $E(3)$- and permutation-equivariant operator with Slater–Koster tight binding as a special case, topological invariants (sheaf cohomology) that count non-bonding orbitals and detect cycle structure, a strict expressivity hierarchy over equivariant MPNNs and CW networks, and an anti-oversmoothing guarantee inherited from non-trivial sheaf diffusion. The framework unifies topological deep learning with equivariant electronic-structure learning and suggests that sheaf cohomology is a natural language for the topology of chemical bonding. The novelty is the formalization and its invariants, not equivariant Hamiltonian prediction per se, which is due to prior work.

Declarations

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Competing interests. The author declares no competing interests.

Author contributions. K. Harish is the sole author and is responsible for all aspects of the work, including conception, theory, implementation, experiments, and writing.

Data and code availability. This study uses no external datasets. All inputs (the Hückel and synthetic tight-binding Hamiltonians and the molecular geometries) are generated analytically by the accompanying code. The code that reproduces every experiment, figure, and reported number is openly available at `https://github.com/krishnatheaverage/equivariant-cellular-sheaves` and archived on Zenodo (DOI: 10.5281/zenodo.20788374).

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